DATA ITEM DESCRIPTION

Title: INLET COMPATIBILITY SUBSTANTIATING DATA REPORT

Number: DI-SESS-82458
Approved Date: 20241202
AMSC Number: F10514
Limitation: N/A
DTIC Applicable: No
GIDEP Applicable: No
Preparing Activity: 11 (AFLCMC/EZFT)
Project Number: SESS-2025-004
Applicable Forms: N/A

Use/Relationship:

The Inlet Compatibility Substantiating Data Report (SDR) will be used to obtain inlet design, performance, and operability information from contractors for evaluating air vehicle airworthiness by government personnel. Inlet compatibility substantiating data includes computational aerodynamics models/results, sub-scale wind tunnel test results, full-scale ground and flight test results, and other related data. The data and results in the inlet compatibility SDR directly contribute to the engine stability audit and are crucial for determining integrated airframe/inlet/engine compatibility.

This Data Item Description (DID) contains the format, content, and intended use information for the data product resulting from the work task described in the contract Statement of Work.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

   The following references are made available at https://www.sae.org/standards/

   SAE ARP1420 Gas Turbine Engine Inlet Flow Distortion Guidelines
   SAE AIR1419 Inlet Total Pressure Distortion Considerations for Gas Turbine Engines
   SAE AIR5686 A Methodology for Assessing Inlet Swirl Distortion
   SAE AIR5687 Inlet/Engine Compatibility – From Model to Full Scale Development
   SAE AIR5866 An Assessment of Planar Waves
   SAE AIR5867 Assessment of the Inlet/Engine Total Temperature Distortion Problem
   SAE ARP6420 Guidelines for Characterization of Gas Turbine Engine Total-Pressure, Planar-Wave, and Total-Temperature Inlet Flow Distortion

   The following references are made available at https://quicksearch.dla.mil/

   DI-SESS-82328 Computational Aerodynamics Analysis Results
   DI-SESS-82361 Acquisition and Sustainment Data Package (ASDP) Technical Report
   MIL-HDBK-831 Preparation of Test Reports
   MIL-HDBK-310 Global Climatic Data for Developing Military Products

2. Format. Unless otherwise specified below, inlet-compatibility substantiating data shall be either in electronic document format, such as portable document format (PDF), the schema required by the acquirer's digital engineering environment, or the supplier's preferred format. Content of electronic documents must be legible to include all text, figures, and text/labels within figures.

3. Content. Inlet compatibility substantiating data shall include details of any physical, computational, or analytical models needed for a third-party assessment of the accuracy of the results. The source of each data item shall be identified. For data determined analytically, the complete set of equations with

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a sample problem shall be presented. For computational aerodynamics models, follow DI-SESS-82328.

Enough information must be provided to understand the propulsion system design and the impact of inlet performance and operability on engine stability throughout the aircraft flight envelope. Unless otherwise specified, all items below should be quantified for Mach numbers, angles of attack, angles of side slip, and Reynolds numbers across the aircraft flight envelope, including static conditions. Information affected by non-standard atmosphere types will be provided for STD, HOT, and COLD days per MIL-HDBK-310.

Inlet compatibility data shall include:

3.1. General data:

3.1.1. Schematic and narrative description of internal aero surfaces with dimensioned detailed scale drawings or a digital model including:

3.1.1.1. Inlet lips, highlight, cowl and forebody geometries.

3.1.1.2. Airframe forebody protrusions that may impact inlet compatibility.

3.1.1.3. Inlet capture area, throat area, and maximum diameter.

3.1.1.4. Diffuser shape, length, and area ratio as a function of fuselage station.

3.1.1.5. Inlet location on aircraft fuselage.

3.1.1.6. Inlet variable geometries, sensing/feedback mechanisms, and scheduling as a function of Mach number and altitude.

3.1.1.7. Secondary flow path geometry and maximum flow rates.

3.1.1.8. Bleed, bypass, auxiliary inlet doors, blow-in doors, ECS, vortex generators, strakes, boundary layer removal schemes and any other passive/active flow control devices.

3.1.2. Integrated airframe/inlet/engine maximum angles of attack and sideslip as a function of Mach number over the full aircraft speed range, including effective sideslip angle for the largest required crosswind at takeoff/static conditions and the largest required in-flight gust load.

3.2. Details on any Computational Aerodynamics (CA) predictions of inlet performance/operability including vision system, sub-scale wind tunnel model, or full-scale ground test. Delivery of CA predictions shall be in accordance with DI-SESS-82328.

3.3. Details of subscale inlet wind tunnel tests:

3.3.1. Model design, model configurations, and model installation in facility.

3.3.2. Facility capabilities and limitations.

3.3.3. An as-executed test matrix.

3.3.4. Differences between model geometry and vision system (or production system) geometry.

3.3.5. Model instrumentation.

3.3.5.1. Forebody and inlet duct pressure measurement types and location.

3.3.5.2. Aerodynamic Instrumentation Plane (AIP) orientation, location, and distribution of any steady-state, dynamic, or swirl probes.

3.3.5.3. Instrumentation manufacturer, type, and dynamic range.

3.3.5.4. List or schematic of locations of failed instrumentation, run numbers where instrumentation was failed, and correction/substitution method.

3.3.5.5. Length/size of pressure tubing and analysis of any associated pressure lags.

3.3.5.6. List of instrumentation names and data file types for use in post-test data processing.

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3.3.6. Mass flow control system, calibration, and instrumentation.
3.3.7. Data acquisition and data processing.
3.3.7.1. Data acquisition system and sampling rate.
3.3.7.2. Low-pass and high-pass filter types and implementation methods. Filters should include ½ to 1-per-rev engine-order filter scaled to wind tunnel geometry, anti-alias filters, A/D filters, and noise filters both analog and digital.

3.4. Details of inlet-engine compatibility flight test:
3.4.1. Inlet, AIP, and engine instrumentation
3.4.1.1. Forebody, inlet duct, fan duct, core duct pressure and temperature measurement location, type, and range
3.4.1.2. AIP number, orientation, location, and distribution of instrumentation. Include any differences between ground and flight test AIP rakes.
3.4.1.3. List of instrumentation names and data file types for use in post-test data processing.
3.4.1.4. List or schematic of locations of failed instrumentation, run numbers where instrumentation is failed, and correction/substitution method.
3.4.1.5. Length/size of pressure tubing and analysis of any associated pressure lags.
3.4.2.Data acquisition and data processing.
3.4.2.1. Data acquisition system and sampling rate.
3.4.2.2. Low-pass and high-pass filter types and implementation methods. Filters should include ½ to 1-per-rev engine-order filter scaled to wind tunnel geometry, anti-alias filters, A/D filters, and noise filters both analog and digital.
3.4.2.3. Any differences between ground/flight methods for distortion descriptors, data processing/screening, and engine stability audit methodology.
3.4.3. Comparison of flight test inlet data with corresponding subscale inlet data at the closest test condition or interpolated conditions from wind tunnel.

3.5. Inlet recovery and distortion analysis results from subscale wind tunnel test, full-scale ground test, and flight test. Distortion descriptors shall be provided in the engine manufacturers index system with equations describing both radial and circumferential directions.
3.5.1. Process of combining AIP steady-state and dynamic total pressure measurements prior to distortion descriptor calculations.
3.5.2. Distortion descriptor calculations.
3.5.3. Details on peak-instantaneous dynamic distortion screening method.
3.5.4. Methodology to combine distortion intensities and engine sensitivities to produce an estimate of the loss in stability pressure ratio of each compression component in accordance with SAE AIR1419 and SAE ARP1420.
3.5.5. Inlet planar wave (subsonic) and buzz (supersonic) regions including the screening parameter, limits, and airflows at which the limit is crossed as a function of Mach number and vehicle attitude.
3.5.6. Supersonic/hypersonic inlet unstart and restart as a function of airflow, Mach number and vehicle attitude.
3.5.7. Ground-to-flight scaling increments for inlet recovery and distortion and description of methodology used in accordance with SAE AIR 5687.
3.5.8. Inlet recovery, steady-state distortion descriptors, peak instantaneous distortion descriptors, and resulting loss in engine stability pressure ratio (for fan and core) as a

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function of airflow, Mach number, angle of attack, angle of sideslip, and Reynolds number.

3.5.9. Effects of icing of inlet recovery, steady-state distortion, and dynamic distortion with/without de-icing systems if the aircraft will fly in icing conditions.

3.5.10. Fan to core distortion transfer characteristics as a function of corrected airflow.

3.5.11. Doghouse plots of steady-state and peak-instantaneous distortion descriptors compared to engine limits for each Mach number regime.

3.5.12. Contour plots of AIP steady-state recoveries and dynamic RMS for relevant worst-case distortion conditions including static conditions and including edges of the doghouse limits in both circumferential, radial directions.

3.5.13. Pressures in the inlet duct or forebody as a function of fuselage station and test conditions.

3.5.14. Hammershock pressure analysis across the flight envelope to ensure engine stall does not result in failed inlet hardware.

3.5.15. Temperature distortion from hot gas ingestion, duct heat transfer, de-icing systems, or reverse inlet secondary flows in accordance with SAE AIR5867.

3.5.16. Swirl distortion descriptors in accordance with SAE AIR 5686 or analysis proving that swirl is mitigated using fan inlet guide vanes.

3.5.17. Deterministic fan and core stability audits showing the impact of peak instantaneous dynamic distortion, temperature distortion, and swirl distortion on fan and core available stability margin.

End of DI-SESS-82458

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